Top-Entry Manhole Lifter: dimension sheet for FreeCAD (prototype, ~300 mm tall)

All values in millimetres. These are PROPOSED starting dimensions, chosen so the parts fit together and print on a normal 220 mm bed. They are not checked in CAD or FEA yet. Adjust them as you model, then update this file.

1. Overall

Item	Value
Overall height (ground to top of gearbox)	300
Foot circle (centre to centre of feet)	Ø 218 (leg tilt 35° from vertical)
Legs	3, at 120° apart
Test cover	Ø 160 x 8, central access hole Ø 28
Lift stroke	30
Light test load	0.5 to 2 kg (confirm)

2. Height stack-up (z = 0 is the ground and the top face of the cover)

z (mm)	What is there
260 to 300	Gearbox housing (40 tall)
140 to 260	Outer telescopic tube (120 long)
140 to 170	Tripod hub (30 tall)
-8 to 0	Test cover
-14 to -8	Folding arms (under the cover)
-54 to -14	Collar, cam and rollers on the inner tube
-54 to 206	Inner tube (260 long, printed in two 130 mm pieces)

3. Parts

ID	Part	Material	Qty	Key dimensions
MH-01a	Tripod hub	PETG	1	Ø 44 x 30; centre bore Ø 22.4; 3 leg sockets 12.4 x 12.4, 18 deep, tilted 35° from vertical
MH-01b	Leg	PETG	3	190 long, 12 x 12 square section, 35° end cuts
MH-02a	Foot pad	TPU	3	Ø 24 x 4, recess 12.4 x 12.4 x 3 for the leg end
MH-02b	Arm contact pad	TPU	4	14 x 10 x 3
MH-02c	Handle grip	TPU	1	Ø 10 x 40, bore Ø 6
MH-03a	Outer tube	PETG	1	OD 22, ID 16.4, 120 long; pin holes Ø 4.2 at z = 230 (cross hole)

MH-03b	Inner tube	PETG	2 pieces	OD 16, ID 10, 130 long each, spigot joint 15 long; 4 pin holes Ø 4.2 at 10 pitch
MH-04	Gearbox housing	PETG	1 set	70 x 60 x 40, wall 3, two halves; wheel bore Ø 10.4, worm bearing holes Ø 6.2
MH-05a	Worm	Nylon	1	Pitch Ø 12, 1 start, module 1.5, 40 long, shaft Ø 6
MH-05b	Worm wheel	Nylon	1	20 teeth, module 1.5, pitch Ø 30, OD 33, 8 thick; ratio 20:1; centre distance 21
MH-06	Folding arm	Nylon	4	55 x 10 x 6; pivot hole Ø 4.2 at 5 from the root; link hole Ø 4.2 at 25 from the root
MH-07a	Sliding collar	Nylon	1	OD 20 x 14 long, bore Ø 16.4; 4 link ears 6 wide with Ø 4.2 holes
MH-07b	Cam	Nylon	1	Ø 24 x 10, bore Ø 16.4
MH-08a	Link	Nylon	4	30 x 8 x 4; holes Ø 4.2, 22 apart
MH-08b	Latch	Nylon	1	30 x 12 x 6 with spring seat
MH-09a	T-handle arm	PETG	1	60 long x 10 x 10
MH-09b	Pull knob	PETG	1	Ø 16 x 12
MH-09c	Status flag	PETG	1	20 x 12 x 2, red on one side, green on the other

MH-10	Test cover	PETG	1	Ø 160 x 8, central hole Ø 28

Bought parts: T8 lead screw (lead 8 mm, 200 long), brass T8 nut (flange Ø 22, 15 long), Ø 4 mm dowel pins x 14 long (8), M4 x 16 bolts and nuts (about 20), Ø 8 steel rollers (2), Ø 4 mm steel rod (1), locking pin Ø 3.5 and spring (1).

4. Clearances (for printed fits)

Fit	Clearance per side
Sliding fits (tubes, collar on tube)	0.2 to 0.3
Pivot holes for Ø 4 pins	0.1 to 0.2 (holes Ø 4.2)
Gear mesh	0.2 backlash allowance
Press fits (leg in hub socket)	0.1

Test-print a pivot joint and a gear pair first and adjust.

5. Printing notes

- Arms, links, collar, cam, worm and wheel: nylon, 4+ walls, 60 to 100% infill, print flat.
- Tubes: PETG, vertical or in two pieces, 3 walls, 40% infill.
- Pads, feet, grip: TPU, slow speed.

6. Checks to do in FreeCAD

1. Arms folded must fit inside the Ø 28 cover hole (folded cluster about Ø 24).
2. Open arm reach: root 8 + 55 = 63 mm radius, inside the cover

radius of 80.

3. Collar travel must swing the arms from folded to flat; adjust the link hole positions in the motion check.
4. Lift stroke: lead 8 mm per wheel turn, so 30 mm is 3.75 wheel turns, or 75 handle turns at 20:1.
5. Total height 300 mm with the gearbox top at z = 300.